

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – INDUSTRY PROBLEM SOLVING TASK

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO4 – Precision (BL4)	Assessment:	Industry Problem solving task
Weightage:	30%	Total Marks:	100
CO4 Statement:	Implement hierarchical memory systems by differentiating cache levels (L1, L2, L3), virtual memory with paging, and advanced memory technologies (DRAM, SRAM, NAND Flash, MRAM, RRAM) based on latency, bandwidth, and throughput characteristics.		
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Section / Batch:	2025	Date:	17-08-2026

Instructions to Students

- Identify the relevant concepts of cache hierarchy, SRAM, DRAM, MRAM, NAND Flash, RRAM, memory latency, bandwidth, and virtual memory paging.
- Analyze the memory-access requirements of smartphones, cloud servers, autonomous vehicles, edge AI devices, and gaming consoles, considering latency, bandwidth, capacity, and power consumption.
- Apply suitable memory-hierarchy techniques such as cache optimization, appropriate memory technology selection, data locality, paging optimization, and hierarchical memory organization.
- Compute performance measures such as memory access time, latency, bandwidth, throughput, speedup, hit/miss rates, and resource or power utilization where applicable.
- Interpret the results by evaluating performance improvements, memory bottlenecks, technology trade-offs, and the suitability of the proposed memory hierarchy for each application.

QUESTIONS

1. A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage in handling these workloads. Propose a memory hierarchy that reduces access latency, improves data locality, and supports high concurrent transaction throughput.
2. A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate the suitability of SRAM, DRAM, and non-volatile memory for storing frequently accessed control data, sensor information, and program code. Design a memory hierarchy that minimizes latency and ensures predictable memory access for real-time operations.
3. A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze how cache size, DRAM capacity, memory bandwidth, and storage performance influence video-processing workloads. Recommend a hierarchical memory organization that improves data transfer rates, reduces processing delays, and supports smooth real-time editing.
4. A weather forecasting system performs complex numerical calculations on massive datasets containing atmospheric temperature, pressure, wind, and humidity measurements. Analyze how cache hierarchy, DRAM bandwidth, and large-capacity memory influence the performance of repeated numerical computations. Propose memory-hierarchy enhancements that improve data locality, reduce memory-access bottlenecks, and increase computational throughput.
5. A smart home gateway manages data from cameras, motion sensors, environmental sensors, and connected appliances while operating continuously with limited energy resources. Compare SRAM, DRAM, Flash, and other suitable memory technologies based on latency, capacity, persistence, and power consumption. Design an efficient memory hierarchy that enables rapid access to frequently used data while minimizing energy consumption and maintaining reliable long-term data storage.

Code of Conduct

I V.Sriram(192524037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: V.Sriram

RUBRICS FOR CALCULATION

Criteria	Weightage(%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

Marks Evaluation:

Q. No	Understanding of Industry Problem (20)	Application of Engineering Concepts (25)	Solution Design (25)	Use of Tools (15)	Analysis (10)	Professionalism(5)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
4	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
5	/ 4	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 25	/ 15	/ 10	/ 5	/ 100

Course Coordinator

Faculty Incharge

QUESTION 1

Problem

A large-scale e-commerce platform experiences slow response times during peak shopping periods when thousands of users simultaneously access product catalogs, customer records, and transaction data. Analyze the roles of cache memory, DRAM, and secondary storage and propose a suitable memory hierarchy.

1. Analysis of the Problem

During peak shopping periods, thousands of users may simultaneously request:

- Product information
- Product prices
- Customer details
- Shopping-cart information
- Transaction records
- Order status

If every request accesses secondary storage directly, the system experiences high latency and reduced throughput. Therefore, a hierarchical memory system is required.

2. Role of Different Memory Levels

Cache Memory

Cache is the fastest memory available close to the processor.

It should store frequently accessed data such as:

- Popular product information
- Frequently searched products
- Prices
- Session information

This reduces repeated access to slower memory.

DRAM

DRAM provides larger capacity than cache and stores:

- Active application data
- Database working sets
- User sessions
- Frequently accessed records

DRAM is slower than cache but much faster than secondary storage.

Secondary Storage

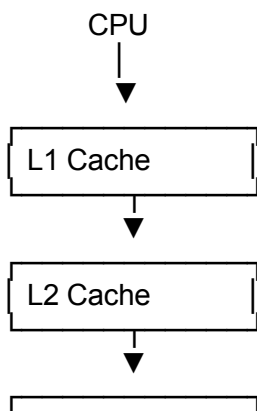
SSD/HDD provides large and persistent storage.

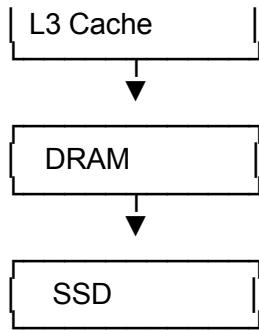
It stores:

- Complete product database
- Customer records
- Historical transactions
- Order information
- Backup data

SSD is preferable to HDD because it provides lower access latency.

3. Proposed Memory Hierarchy





4. Data Locality

Two important types of locality can be used.

Temporal Locality

Recently accessed data is likely to be accessed again.

Example:

A customer repeatedly views the same product.

The product information can remain in cache.

Spatial Locality

Data located close to recently accessed data is likely to be accessed.

Example:

Product information and its price can be stored close together.

5. Performance Calculation

Assume:

- Cache access time = 1 ns
- DRAM access time = 80 ns
- SSD access time = 100,000 ns
- Cache hit rate = 95%

For a simplified two-level calculation:

Average Access Time

= Hit rate × Cache time

+ Miss rate × DRAM time

Therefore:

$= 0.95 \times 1 + 0.05 \times 80$

$= 0.95 + 4$

$= 4.95 \text{ ns}$

Without cache:

Access time = 80 ns

Approximate improvement:

$80 / 4.95 \approx 16.16$

So caching can provide approximately 16× lower average access time in this simplified example.

6. Recommended Solution

The e-commerce platform should use:

- Large L3 cache
- Sufficient DRAM
- High-speed NVMe SSD
- Database caching
- Frequently accessed data in cache
- Data replication for high availability
- Efficient memory allocation

Result

The proposed hierarchical memory system reduces repeated slow-storage accesses, improves data locality,

decreases average memory-access latency, and supports high concurrent transaction throughput.

QUESTION 2

Problem

A robotic manufacturing system must process camera images, sensor readings, and control instructions with strict real-time deadlines. Evaluate SRAM, DRAM, and non-volatile memory and design a hierarchy that minimizes latency and provides predictable access.

1. Analysis of the Problem

A manufacturing robot continuously receives:

- Camera images
- Temperature readings
- Position information
- Motor feedback
- Sensor measurements
- Control instructions

The system must respond within strict deadlines.

Therefore, **predictable and low memory latency** is more important than simply having large storage capacity.

2. SRAM

SRAM is very fast and does not require refresh operations.

Advantages

- Very low latency
- High speed
- Predictable access
- Suitable for frequently accessed control data

Disadvantages

- Expensive
- Low storage density
- Higher cost per bit

Suitable for

- CPU cache
- Critical control variables
- Real-time buffers

3. DRAM

DRAM provides larger capacity than SRAM.

Advantages

- Higher capacity
- Lower cost per bit than SRAM
- Suitable for large sensor data

Disadvantages

- Higher latency than SRAM
- Requires refresh
- Access time is less predictable

Suitable for

- Camera frames
- Sensor data buffers
- Operating-system memory

4. Non-Volatile Memory

Examples include:

- NAND Flash
- EEPROM
- MRAM

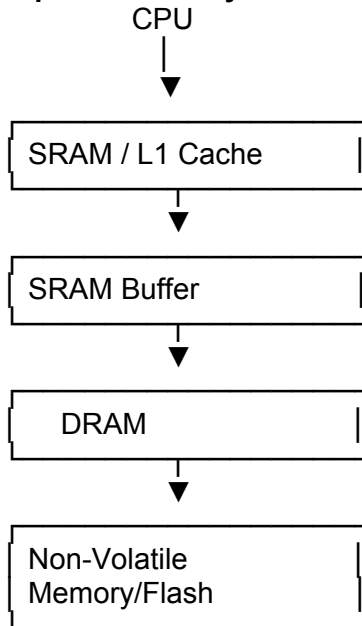
- RRAM

Non-volatile memory retains information even after power is removed.

Suitable for:

- Program code
- Configuration
- Calibration data
- System parameters

5. Proposed Memory Hierarchy



6. Data Placement

Data	Recommended Memory
Critical control variables	SRAM
Motor control data	SRAM
Sensor readings	SRAM/DRAM
Camera frames	DRAM
Program code	Flash
Configuration	Non-volatile memory
Calibration data	Non-volatile memory

7. Latency Analysis

Assume:

SRAM = 2 ns

DRAM = 80 ns

Flash = 100,000 ns

For critical control data, SRAM is clearly preferred.

Relative to SRAM:

$\text{DRAM} / \text{SRAM} = 80 / 2 = 40$

DRAM access is approximately **40× slower** in this example.

Therefore, placing time-critical control data in SRAM reduces the possibility of missing real-time deadlines.

8. Real-Time Optimization

The system should:

1. Keep critical variables in SRAM.
2. Use DRAM for large image buffers.
3. Keep permanent program code in Flash.
4. Use DMA for transferring sensor/camera data.
5. Use cache carefully for predictable access.

6. Avoid unnecessary paging during real-time operation.

Result

SRAM is the most suitable technology for time-critical control data, DRAM is suitable for large sensor and image buffers, and non-volatile memory is suitable for program code and permanent configuration. This hierarchy minimizes latency and provides more predictable real-time operation.

QUESTION 3

Problem

A video-editing workstation experiences delays while processing multiple streams of high-resolution 4K and 8K video. Analyze cache size, DRAM capacity, memory bandwidth, and storage performance and recommend a hierarchy.

1. Analysis of the Problem

4K and 8K video processing requires large amounts of data to be moved between:

- Storage
- DRAM
- CPU
- GPU
- Cache
- Video-processing units

If the memory bandwidth is insufficient, the processor/GPU may remain idle while waiting for data.

2. Importance of Cache

Cache stores frequently accessed instructions and data close to the processor.

A larger cache can reduce accesses to DRAM.

However, increasing cache size indefinitely is not always beneficial because:

- It increases cost.
- Larger caches may have longer access times.
- Power consumption increases.

Therefore, a balanced cache hierarchy is required.

3. Importance of DRAM Capacity

Video editing requires large working datasets.

Insufficient DRAM can cause data to be moved to storage.

This is much slower than DRAM and causes:

DRAM shortage

↓

More storage access

↓

Higher latency

↓

Lower editing performance

Therefore, high-capacity DRAM is important.

4. Memory Bandwidth

Memory bandwidth determines how much data can be transferred per second.

Assume:

Memory bus width = 256 bits

Clock frequency = 3.2 GHz

256 bits:

$256 / 8 = 32$ bytes

Theoretical bandwidth:

32×3.2 billion

= 102.4 GB/s

Thus, the theoretical memory bandwidth is:

102.4 GB/s

Actual bandwidth will be lower because of system overhead and memory-access patterns.

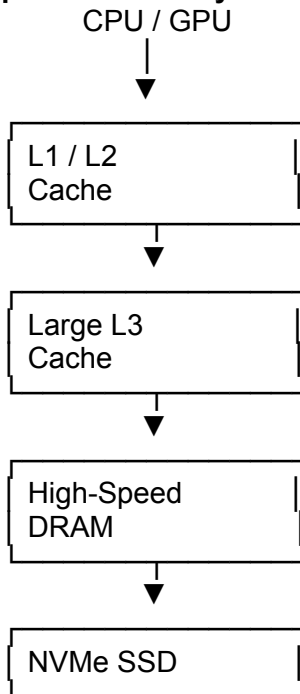
5. Storage Performance

Video editing requires fast storage for:

- Raw video
- Project files
- Rendered files
- Cache files
- Temporary files

NVMe SSD is preferable to HDD because of its much higher throughput and lower latency.

6. Proposed Hierarchy



7. Recommended Optimization

Cache

Use a sufficiently large multi-level cache to exploit temporal and spatial locality.

DRAM

Use high-capacity, high-bandwidth memory.

Storage

Use NVMe SSD for active projects and media files.

GPU

Use high-bandwidth graphics memory for video processing.

Data Transfer

Use DMA to transfer large video blocks without excessive CPU involvement.

8. Performance Comparison

Suppose:

System A bandwidth = 50 GB/s

System B bandwidth = 100 GB/s

For a 200 GB dataset:

System A:

Time = $200 / 50$

= 4 seconds

System B:

Time = $200 / 100$

= 2 seconds

Speedup:

$$4 / 2 = 2\times$$

Therefore, doubling bandwidth can approximately halve the idealized transfer time for a bandwidth-bound workload.

Result

A combination of large cache capacity, high-capacity/high-bandwidth DRAM, GPU memory, and NVMe SSD significantly improves high-resolution video processing. The proposed hierarchy reduces data-transfer delays and supports smoother 4K and 8K editing.

QUESTION 4

Problem

A weather forecasting system performs complex numerical calculations on massive atmospheric datasets containing temperature, pressure, wind, and humidity measurements. Analyze cache hierarchy, DRAM bandwidth, and large-capacity memory and propose improvements.

1. Analysis of the Problem

Weather forecasting involves repeated numerical operations on very large datasets.

Typical variables include:

- Temperature
- Pressure
- Wind velocity
- Humidity
- Atmospheric density

The same data may be accessed repeatedly during numerical calculations.

Therefore, **data locality and memory bandwidth** are extremely important.

2. Role of Cache Hierarchy

L1 Cache

Stores the most frequently used data and instructions.

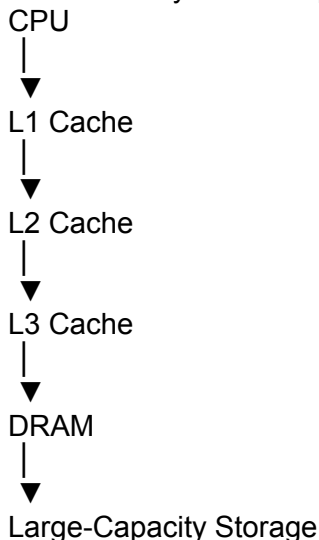
L2 Cache

Provides greater capacity while remaining relatively fast.

L3 Cache

Provides a larger shared cache and reduces DRAM accesses.

The hierarchy can be represented as:



3. Importance of Data Locality

Weather calculations often operate on neighboring grid points.

For example:

Grid point A



Neighboring grid points



Repeated calculations

If neighboring data is stored contiguously, spatial locality improves.

4. Cache Blocking

A large dataset should be divided into smaller blocks.

Instead of processing:

Entire Dataset

process:

Block 1 → Block 2 → Block 3 → ...

A block should fit effectively within the available cache.

This reduces repeated DRAM accesses.

5. DRAM Bandwidth

Assume:

DRAM bandwidth = 200 GB/s

Dataset transfer = 1 TB

Since:

1 TB = 1000 GB

Ideal transfer time:

$1000 / 200$

= 5 seconds

If bandwidth is increased to 400 GB/s:

$1000 / 400$

= 2.5 seconds

Ideal bandwidth-limited transfer time is therefore reduced by half.

6. Memory Bottleneck

If the CPU can perform calculations faster than data can be supplied, the system becomes **memory-bound**.

CPU computation



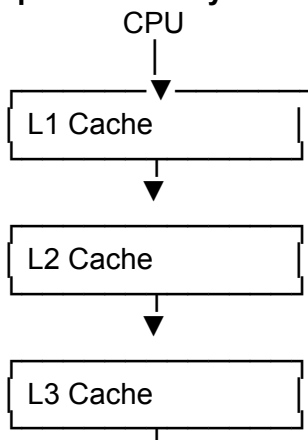
Waiting for data

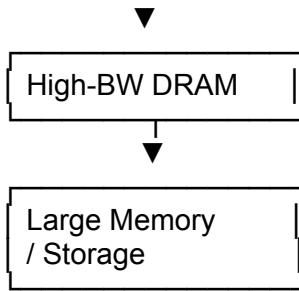


DRAM bandwidth bottleneck

This reduces computational throughput.

7. Proposed Memory Hierarchy





8. Recommended Enhancements

1. Increase DRAM bandwidth.
2. Use large shared L3 cache.
3. Apply cache blocking.
4. Store neighboring grid data contiguously.
5. Use prefetching where appropriate.
6. Use parallel processing.
7. Reduce unnecessary data movement.
8. Use high-capacity memory for massive datasets.

9. Performance Analysis

Technique	Effect
Larger L1/L2 cache	Fewer cache misses
Larger L3 cache	Less DRAM traffic
Higher DRAM bandwidth	Faster data supply
Cache blocking	Better locality
Prefetching	Hides memory latency
Parallel memory access	Higher throughput

Result

The weather forecasting system benefits from a multi-level cache hierarchy, high-bandwidth DRAM, cache blocking, prefetching, and efficient data layout. These techniques improve data locality, reduce memory bottlenecks, and increase numerical-computation throughput.

QUESTION 5

Problem

A smart home gateway continuously manages cameras, motion sensors, environmental sensors, and connected appliances while operating under limited energy resources. Compare SRAM, DRAM, Flash, and other suitable technologies and design an energy-efficient memory hierarchy.

1. Analysis of the Problem

A smart home gateway receives data continuously from:

- Security cameras
- Motion sensors
- Temperature sensors
- Humidity sensors
- Smart appliances

The system must provide fast access while consuming limited energy.

Therefore, memory selection should consider:

- Latency
- Capacity
- Persistence
- Power consumption

- Reliability
- Cost

2. SRAM

Characteristics

- Very low latency
- High speed
- Volatile
- Small capacity
- Relatively expensive

Suitable for

- Frequently accessed variables
- Processor cache
- Temporary control information

3. DRAM

Characteristics

- Higher capacity than SRAM
- Volatile
- Moderate latency
- Lower cost per bit
- Requires refresh

Suitable for

- Sensor buffers
- Camera processing
- Application working memory

4. Flash

Characteristics

- Non-volatile
- High capacity
- Retains data without power
- Lower cost per bit
- Slower than SRAM/DRAM

Suitable for

- Operating system
- Application code
- Long-term logs
- Configuration data

5. MRAM/RRAM

Advanced non-volatile technologies such as **MRAM** and **RRAM** can provide useful combinations of persistence, speed, endurance, and energy efficiency depending on implementation.

They can be considered for:

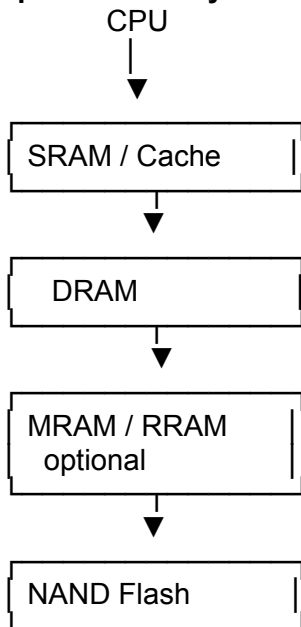
- Frequently updated persistent information
- Configuration
- Fast non-volatile storage

The assignment specifically identifies SRAM, DRAM, NAND Flash, MRAM, and RRAM as memory technologies to differentiate based on latency, bandwidth, and throughput.

6. Comparison

Memory	Latency	Capacity	Persistent?	Power/Use
SRAM	Very Low	Low	No	High speed
DRAM	Low/Medium	High	No	Requires refresh
Flash	High	Very High	Yes	Good for storage
MRAM	Low	Medium	Yes	Low standby potential
RRAM	Low/Medium	High potential	Yes	Potentially energy efficient

7. Proposed Memory Hierarchy



8. Data Placement

Data

Frequently accessed sensor values

Active camera processing buffers

Application working data

Persistent configuration

Operating system

Long-term sensor logs

Critical persistent parameters

Memory

SRAM/Cache

DRAM

DRAM

Flash/MRAM/RRAM

Flash

Flash

MRAM/RRAM

9. Energy Optimization

1. Keep frequently accessed data in cache

This reduces repeated DRAM accesses.

2. Use sleep modes

Unused memory components can enter low-power states.

3. Reduce unnecessary data movement

Moving data consumes energy.

4. Buffer sensor data

Batching sensor data can reduce frequent memory accesses.

5. Use Flash for inactive data

Inactive data does not need to occupy high-speed volatile memory.

10. Performance Example

Suppose a sensor application generates:

100 MB of data

If effective DRAM bandwidth is:

10 GB/s

Ideal transfer time:

$100 \text{ MB} / 10 \text{ GB/s}$

$\approx 0.01 \text{ seconds}$

Therefore:

Transfer time $\approx 10 \text{ ms}$

If only 5 GB/s is available:

100 MB / 5 GB/s

≈ 20 ms

Thus, doubling bandwidth reduces ideal transfer time by approximately half.

11. Recommended Design

For the smart home gateway:

Fast access

↓
SRAM / Cache

↓
Working data

↓
DRAM

↓
Persistent frequently accessed data

↓
MRAM / RRAM

↓
Long-term storage

↓
NAND Flash

This provides:

- Fast access to frequently used data
- Adequate capacity for camera/sensor processing
- Persistent long-term storage
- Reduced unnecessary power consumption

Result

An energy-efficient smart home gateway should use SRAM/cache for frequently accessed data, DRAM for active working data, and Flash for high-capacity persistent storage. MRAM or RRAM can be considered for suitable persistent, frequently accessed information. This hierarchical organization balances latency, capacity, persistence, reliability, and energy consumption.